

# AGNI-NETRA

## AI Geospatial Network for Industrial Thermal Risk & Anomaly Analysis

INTELLIGENCE DOSSIER • Generated on 2026-09-12 06:52 UTC • DECISION SUPPORT SYSTEM

|                                            |                                                   |
|--------------------------------------------|---------------------------------------------------|
| Event Code: EVT-827                        | Classification: Industrial Fire (97.1%)           |
| Location: 22.35420°N, 69.86440°E (Gujarat) | AGNI-NETRA Risk Level: <b>CRITICAL</b> (80.3/100) |
| Active Detections: 6 observations          | Peak Radiative Power (FRP): 285.0 MW              |
| First Detected: 2026-08-30 15:26:52        | Last Detected: 2026-08-30 15:26:52                |

### 1. Explainable AI (SHAP) Attribution

Evaluated via candidate XGBoost v3.0 model with TreeExplainer SHAP attribution.

| Feature Name       | Value              | Shapley Impact | Direction          |
|--------------------|--------------------|----------------|--------------------|
| dist_to_facility_m | 181.91402943653523 | +0.420         | Supports Class (+) |

### 2. Geospatial Facility Association

|                                              |                                           |
|----------------------------------------------|-------------------------------------------|
| Associated Facility: None (Uncataloged Site) | Facility Type: N/A                        |
| Facility Status: KNOWN                       | Distance to Boundary: 181.9 m             |
| Land Cover Context: Industrial               | Source Provenance: OSM / Satellite Survey |

### 3. Risk Evaluation & Drivers

- Severe radiative heat output (Peak FRP: 285.0 MW)
- Current FRP (285.0 MW) is statistically critical (+6.4σ above historical baseline mean of 125.0 MW).
- Chronic persistent thermal source with continuous multiday emissions
- Direct correlation with high-hazard industrial facility boundary

### 4. Model Governance & Data Provenance

|                                                                        |                                                          |
|------------------------------------------------------------------------|----------------------------------------------------------|
| Model Version: XGBoost V3 (xgb-v3.0-real-candidate • Platt-Calibrated) | Telemetry Source: NASA FIRMS S-NPP/VIIRS 375m NRT        |
| Spatial Context Engine: PostGIS 3.4 Multi-Layer Topology               | Verification Status: ACTIVE (Pending Human Verification) |
| Automated Dispatch: DISABLED / GATED SAFE (Human Decision Support)     | Dossier Generated: 2026-09-12 06:52:05 UTC               |

**LEGAL & SCIENTIFIC DISCLAIMER:** This document contains automated analytical intelligence derived from satellite remote sensing observations and machine learning models. Predictions reflect statistical probabilities and do not constitute certified ground truth until verified by authorized domain personnel.